

Assessment 2 Individual Workbook

Section 1: Personal contributions to the team design portfolio

Personal Contribution

In this project, I worked as the **Team Leader**, **Game Designer**, and **Scrum Coordinator**. My main contribution was designing and developing three comparison game levels and integrating the game interfaces, sketches, and design materials into the shared **Figma portfolio**. I collaborated with team members to organise and update the Figma document so that the overall design remained consistent. As the Scrum Coordinator, I monitored each member's progress, followed up on incomplete tasks, and helped ensure that the planned work was completed within the **Sprint Pilot**. I also supported team discussions, recorded feedback from the **Scrum Review**, and contributed to the **Sprint Retrospective** by identifying problems, lessons learned, and possible improvements for future work.

User story 1:

Stub:Animal Size Comparison

User story: As a kindergarten student, I want to compare cartoon animals using greater than, less than, and equal signs so that I can learn animal names and understand basic size comparison in English.

Acceptance criteria:

- 1.The screen displays clear cartoon animal pairs such as tiger, rabbit, dog, panda, pig, and snail.
- 2.Each animal pair has a suitable comparison sign for matching: greater than, less than, or equal to.
- 3.Correct matches play a short sound and show positive feedback.

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/Untitled?node-id=1-5&t=RB35blqDcUiyjX0D-1>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14377&qid=8654&lid=5375&sid=2938>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	kindergart en student	an animal size comparison game level	I can learn common animal names in English
2	kindergart en student	compare cartoon animal pairs using <, > and = signs	I can understand which animal is bigger, smaller, or the same size
3	kindergart en student	hear feedback after choosing the correct comparison sign	I can know whether my answer is correct
4	kindergart en student	receive simple positive feedback after answering correctly	I can feel encouraged to continue learning

Feedback

1. The game type should be more closely related to the chosen topic.
2. Some details, such as the layout, instructions, and interactions, still need to be improved.
3. Suitable and age-appropriate English words should also be prepared for each game level.

User story 2:

Stub:Item Size Comparison

User story: As a kindergarten student, I want to compare cartoon items using greater than, less than, and equal signs so that I can learn common item names and understand basic size comparison in English.

Acceptance criteria:

- 1.The screen displays clear cartoon item pairs such as coin, ore, branch, maple leaf, glove, cape, keys, backpack, and wallet.
- 2.Each item pair has a suitable comparison sign for matching: greater than, less than, or equal to.
- 3.Correct matches play a short sound and show positive feedback.

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/Untitled?node-id=1-5&t=RB35blqDcUiyjX0D-1>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14377&qid=8660&lid=5375&sid=2938>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	kindergart en student	an item size comparison game level	I can learn common item names in English
2	kindergart en student	compare cartoon item pairs using <, > and = signs	I can understand which item is bigger, smaller, or the same size
3	kindergart en student	hear feedback after choosing the correct comparison sign	I can know whether my answer is correct
4	kindergart en student	receive simple positive feedback after answering correctly	I can feel encouraged to continue learning

Feedback

1. The item comparison activity should show a clearer connection between the game mechanics and English vocabulary learning.
2. The interface, comparison symbols, and instructions could be refined to make the gameplay easier to understand.
3. Simple and familiar item words should be selected to ensure that the level is suitable for kindergarten learners.

User story 3:

Stub: Food Size Comparison

User story: As a kindergarten student, I want to compare cartoon foods using greater than, less than, and equal signs so that I can learn common food names and understand basic size comparison in English.

Acceptance criteria:

1. The screen displays clear cartoon food pairs such as cake, milk, strawberry, banana, blueberry, apple, coconut, lemon, and pineapple.

2. Each food pair has a suitable comparison sign for matching: greater than, less than, or equal to.

3. Correct matches play a short sound and show positive feedback.

Story size: M

URL to prototype (Figma prototype design):

<https://www.figma.com/board/wcrZx8euXC17OJRNfkbkiT/Untitled?node-id=1-5&t=RB35blqDcUiYjX0D-1>

URL to the game:

<https://www.kopo.com/g/game.php?gno=14377&qid=8661&lid=5375&sid=2938>

Sprint Pilot Backlog

ID	As a	I want to able to play	So that
1	kindergarten student	a food size comparison game level	I can learn common food names in English
2	kindergarten student	compare cartoon food pairs using <, > and = signs	I can understand which food is bigger, smaller, or the same size
3	kindergarten student	hear feedback after choosing the correct comparison sign	I can know whether my answer is correct
4	kindergarten student	receive simple positive feedback after answering correctly	I can feel encouraged to continue learning

Feedback

1. The food comparison level should connect the size-comparison task more clearly with English vocabulary learning.
2. The instructions, symbols, and image layout should remain simple and consistent for kindergarten students.
3. Familiar food words and colourful shapes should be used to make the activity easier and more engaging.

Section 2: Rationale behind key design decisions you were involved in

The main goal of this project was to design three simple **English educational game levels** for kindergarten students using the KOPO game designer. The three deliverables were **Animal Size Comparison**, **Item Size Comparison**, and **Food Size Comparison**. The client expected the games to be **engaging, age-appropriate**, visually clear, and able to support basic English vocabulary learning. However, the work had to be completed within a limited **Sprint Pilot**, and KOPO imposed fixed interaction functions and image dimensions. Therefore, I assumed that young learners would benefit most from familiar words, cartoon images, short instructions, and one consistent comparison mechanic.

I applied an **Agile Development** approach by converting the client's broad requirements into three focused **User Stories**. Each story followed the structure: "As a kindergarten student, I want to compare cartoon objects so that I can learn English words and basic size comparison." This format kept the requirements brief, understandable, and connected to user value. Clear and self-contained user stories can reduce ambiguity and support communication between stakeholders and designers (Lucassen et al., 2016). I also created measurable **Acceptance Criteria**, including five image pairs, readable English labels, suitable <, > and = signs, sound feedback, and positive responses. These criteria helped transform general expectations such as "fun" and "educational" into observable design outcomes.

The three themes were selected because they included familiar and visually recognisable vocabulary. Animals such as dogs, rabbits, and pandas are easy to distinguish, while common items and foods introduce useful English nouns. Connecting pictures directly with written words can support vocabulary recognition and recall (Yip & Kwan, 2006). Educational games may also strengthen second-language vocabulary acquisition when the learning objective remains clear and is not replaced by entertainment alone (Calvo-Ferrer, 2017). For this reason, every picture included an English label, while the comparison task required learners to examine both the object and its name.

The same **Interaction Design** and **Grid Layout** were used across all three levels. Each row contained two images, their labels, and one comparison sign. The **Similarity Principle** was applied through consistent card sizes, fonts, and symbol styles, while the **Proximity Principle** grouped each label closely with its corresponding image. This reduced unnecessary visual complexity and made the relationship between the objects and answers easier to understand. An alternative was to use different mechanics for each level, but this would have increased the learner's **Cognitive Load** and reduced interface consistency.

Colourful cartoon assets, large symbols, short titles, and immediate feedback were used to improve **Engagement**. Effective **Game-Based Learning** depends on cognitive, emotional, motivational, and behavioural involvement rather than decoration alone (Plass et al., 2015). Serious games can improve learning and motivation, particularly when they provide repeated practice and clear instructional support (Wouters et al., 2013). Therefore, the visual design was combined with simple learning tasks and immediate confirmation after correct answers.

I used **Software Prototyping** to progressively refine the interface. A hand-drawn **Low-Fidelity Prototype** was first created to explore layout, content grouping, and symbol positions. A digital **Medium-Fidelity Prototype** was then developed in Figma to present a clearer interface structure. Prototypes allow designers to test selected ideas without developing the complete system (Lim et al., 2008), while different fidelity levels support evaluation of layout and interaction before detailed visual refinement (McCurdy et al., 2006). Low-fidelity prototypes can also reveal important **Usability Problems** early and at relatively low cost (Virzi et al., 1996).

A major **Design Trade-off** was balancing visual appeal with clarity. Decorative backgrounds increased interest but could distract from the central comparison task. Therefore, the learning content was placed inside a separate panel. Some MapleStory-inspired assets also had inconsistent sizes and quality, so they were resized and simplified. Although this reduced visual authenticity, it improved **Consistency**, **Legibility**, and overall **Usability**, which were more important for the target learners.

Section 3: Reflects on feedback, iteration, constraints, and design trade-offs

The design was revised during the **Sprint Pilot** because the first versions did not fully demonstrate the intended English-learning value. Feedback showed that the **layout**, **instructions**, **interactions**, and vocabulary required improvement. The client also requested five comparison pairs, five operators, clearer “Compare...” instructions, and more child-friendly shapes. This showed that completing visual assets alone did not satisfy the **User Stories** or **Acceptance Criteria**; the design also needed to deliver clear learning value.

Initially, I focused mainly on colourful MapleStory-inspired assets and attractive backgrounds. However, the feedback made me realise that visual appeal does not automatically create effective **Game-Based Learning**. Games support learning when their mechanics promote cognitive, behavioural, emotional, and social

engagement with the subject matter. I therefore revised the titles, added five complete comparison rows to each level, used familiar English labels, and included immediate sound and visual feedback. Vocabulary games are more effective when words are integrated into meaningful activities, while game-based second-language learning requires a clear instructional purpose (Calvo-Ferrer, 2017; Plass et al., 2015; Yip & Kwan, 2006).

The **Scrum Review** was important because it enabled **Transparency**, **Inspection**, and **Adaptation** before the designs were treated as complete. Without review, the team could have followed a **Scrum anti-pattern** by completing backlog tasks without checking whether the increment delivered actual user value. Another risk was treating the original plan as fixed despite new evidence. Instead, feedback was used to revise the backlog and redesign the levels. Agile team effectiveness depends strongly on coordination, shared understanding, leadership, and communication (Moe et al., 2010).

The use of **Low-Fidelity Prototypes**, **Medium-Fidelity Prototypes**, and final KOPO screens supported **Iterative Design**. Paper sketches helped me adjust the structure quickly, while Figma clarified the **Grid Layout**, spacing, **Visual Hierarchy**, and control positions. Prototypes act as design filters that allow selected interface features to be explored before full development (Lim et al., 2008). Different fidelity levels also help direct limited time and effort toward uncertain design issues (McCurdy et al., 2006). Low-fidelity prototypes can identify important **Usability Problems** without requiring a complete product (Virzi et al., 1996).

Performance was measured against the revised **Acceptance Criteria**. The final levels contained five labelled pairs, five operators, consistent instructions, and repeated **Interaction Patterns**. Therefore, the main **Functional Requirements** were achieved. However, a major **Constraint** was the absence of formal testing with kindergarten students. Adult approval does not prove that children will understand or enjoy an interface, so children's opinions should be collected directly through suitable **Usability Evaluation** methods (Read, 2008).

The main **Design Trade-off** was between visual richness and **Usability**. Decorative backgrounds increased interest but could distract from the learning task. I therefore placed the comparisons inside a separate central panel and used consistent cards and symbols. KOPO's fixed image sizes and limited interactions also reduced **Design Flexibility**. My key takeaway is that **Educational Game Design** should prioritise **Clarity**, **Validation**, and **Learning Value** before decoration. In future, I would conduct short **User Testing** sessions, record task success and confusion points, refine the **Product Backlog** after each review, and use the **Sprint Retrospective** to improve both the product and the team process.

Reference

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